Examiner only

Answer **all** questions.

1. (a) Rock salt is a mixture of salt and sand. Crystals of pure salt can be obtained from rock salt.

A–E are the steps in the method used but they are in the wrong order.

A Add water to a sample of rock salt in a beaker and stir

B Heat the solution to evaporate some of the water

C Grind the rock salt into a fine powder

D Filter the mixture to separate the sand from the salt solution

E Leave the saturated solution in a warm place for a few days so that crystals of salt form

Put the steps in the correct order. The first step is already included. [2]

C				
first step				

(b) A student was asked to investigate the dyes present in an orange sweet.

The student carried out the following method. There are two errors in the method.

- Draw a line using a ruler and pen on chromatography paper.
- Place a sample of the orange colour on the line.
- Stand the chromatography paper in a beaker and add enough water to just cover the sample.
- Leave the paper to stand until the water rises to the top of the paper.

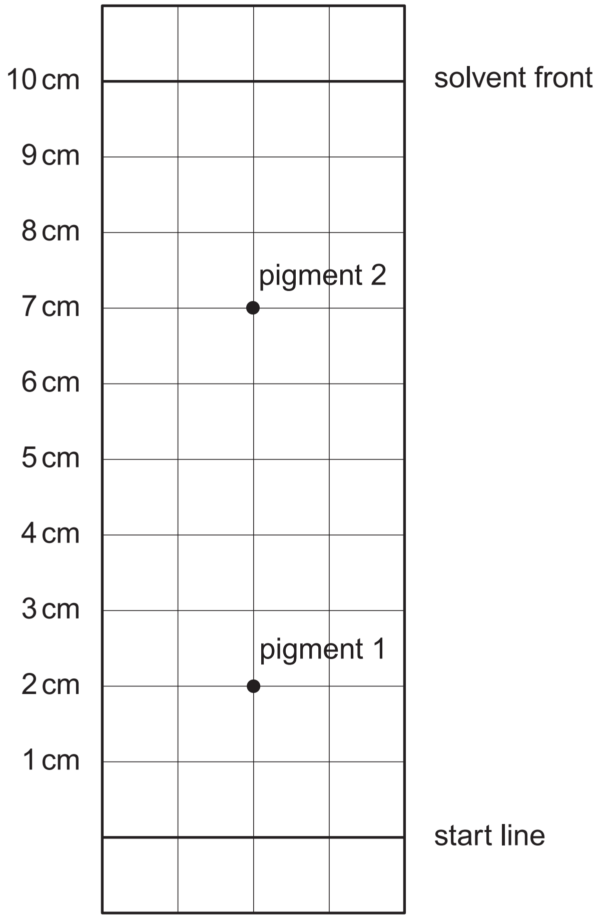
(i) State the **two** errors in the method. [2]

1. ....

2. ....

Examiner only

(ii) Another student used a correct method and obtained the chromatogram below.



Use the formula to calculate the R_f value for pigment 2. [2]

$$R_f = \frac{\text{distance travelled by the pigment}}{\text{distance travelled by the solvent front}}$$

R_f = ...

3430U201 03

6